

script.r

Harshal

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```
library(readxl)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(jsonlite)
library(DBI)
library(RSQLite)
```

```
# import the file into env
raw <- read_excel("Online Retail.xlsx", sheet = 1)
```

```
# splits the sheets
transactions <- raw %>%
  select(InvoiceNo, StockCode, CustomerID, Quantity, InvoiceDate)
```

```
products <- raw %>%
  distinct(StockCode, .keep_all = TRUE) %>%
  select(StockCode, Description, UnitPrice)
```

```
customers <- raw %>%
  distinct(CustomerID, .keep_all = TRUE) %>%
  select(CustomerID, Country)
```

```
# writes the contents of the sheets into separate files (csv for transactions, json for products and excel for customers)
write_csv(transactions, "transactions.csv", row.names = FALSE)
write_json(products, "products.json")
write_xlsx(customers, "customers.xlsx")
```

```
transactions <- read_csv("transactions.csv", stringsAsFactors = FALSE)
products <- fromJSON("products.json")
```

```

customers <- read_excel("customers.xlsx")

# cleaning each dataset
transactions <- transactions %>%
  filter(!is.na(CustomerID), !is.na(InvoiceNo), !is.na(StockCode)) %>%
  distinct() %>%
  filter(Quantity > 0)

products <- products %>%
  filter(!is.na(StockCode), !is.na(UnitPrice)) %>%
  distinct(StockCode, .keep_all = TRUE) %>%
  filter(UnitPrice > 0)

customers <- customers %>%
  filter(!is.na(CustomerID)) %>%
  distinct(CustomerID, .keep_all = TRUE)

# only keeps related data. in case some data has no match in other tables, it will be gone
integrated <- transactions %>%
  inner_join(products, by = "StockCode") %>%
  left_join(customers, by = "CustomerID") %>%
  mutate(Revenue = Quantity * UnitPrice)

dim(integrated)

```

```
## [1] 387877      9
```

```
sum(is.na(integrated$Country))
```

```
## [1] 0
```

```

# statistics and grouping
total_revenue <- sum(integrated$Revenue)

top5_products <- integrated %>%
  group_by(StockCode, Description) %>%
  summarise(Revenue = sum(Revenue), .groups = "drop") %>%
  arrange(desc(Revenue)) %>%
  head(5)

top5_countries <- integrated %>%
  group_by(Country) %>%
  summarise(Revenue = sum(Revenue), .groups = "drop") %>%
  arrange(desc(Revenue)) %>%
  head(5)

customer_value <- integrated %>%
  group_by(CustomerID) %>%
  summarise(TotalValue = sum(Revenue), .groups = "drop")

top5_customers <- customer_value %>%
  arrange(desc(TotalValue)) %>%

```

```

head(5)

customer_value <- customer_value %>%
  mutate(Segment = case_when(
    TotalValue < 500 ~ "Low Value",
    TotalValue < 2000 ~ "Medium Value",
    TotalValue < 5000 ~ "High Value",
    TRUE ~ "Premium"
  ))

country_revenue <- integrated %>%
  group_by(Country) %>%
  summarise(Revenue = sum(Revenue), .groups = "drop") %>%
  arrange(desc(Revenue))

high_market <- country_revenue %>% slice(1)
low_market <- country_revenue %>% slice(n())

# sqlite storage tasks for storing the above analysis and data
con <- dbConnect(RSQLite::SQLite(), "retail_sales.db")
dbWriteTable(con, "retail_sales", integrated, overwrite = TRUE)

query1 <- dbGetQuery(con, "
  SELECT CustomerID, SUM(Revenue) AS TotalRevenue
  FROM retail_sales
  GROUP BY CustomerID
  ORDER BY TotalRevenue DESC
  LIMIT 5
")

query2 <- dbGetQuery(con, "
  SELECT Country, SUM(Revenue) AS TotalRevenue
  FROM retail_sales
  GROUP BY Country
  ORDER BY TotalRevenue DESC
")

dbDisconnect(con)

# console output for analysis
print(total_revenue)

```

```
## [1] 10752840
```

```
print(top5_products)
```

```
## # A tibble: 5 x 3
##   StockCode Description      Revenue
##   <chr>      <chr>      <dbl>
## 1 23843      PAPER CRAFT , LITTLE BIRDIE 168470.
## 2 47566      PARTY BUNTING               142438.
## 3 22423      REGENCY CAKESTAND 3 TIER     135605.
```

```
## 4 85123A    WHITE HANGING HEART T-LIGHT HOLDER  93746.
## 5 23166     MEDIUM CERAMIC TOP STORAGE JAR           81033.
```

```
print(top5_countries)
```

```
## # A tibble: 5 x 2
##   Country      Revenue
##   <chr>         <dbl>
## 1 United Kingdom 8861857.
## 2 Netherlands   363884.
## 3 EIRE           331660.
## 4 Germany        263819.
## 5 France         226976.
```

```
print(top5_customers)
```

```
## # A tibble: 5 x 2
##   CustomerID TotalValue
##         <dbl>     <dbl>
## 1      18102    408760.
## 2     14646    357531.
## 3     17450    186038.
## 4     14911    182691.
## 5     16446    168472.
```

```
print(high_market)
```

```
## # A tibble: 1 x 2
##   Country      Revenue
##   <chr>         <dbl>
## 1 United Kingdom 8861857.
```

```
print(low_market)
```

```
## # A tibble: 1 x 2
##   Country      Revenue
##   <chr>         <dbl>
## 1 Saudi Arabia    181.
```

```
print(query1)
```

```
##   CustomerID TotalRevenue
## 1      18102    408760.0
## 2     14646    357531.1
## 3     17450    186038.0
## 4     14911    182690.5
## 5     16446    168472.5
```

```
print(query2)
```

```
##          Country TotalRevenue
## 1      United Kingdom  8861857.13
## 2      Netherlands  363884.48
## 3          EIRE  331660.17
## 4      Germany  263818.97
## 5      France  226975.60
## 6      Australia  173918.61
## 7      Spain  67426.09
## 8      Switzerland  66619.97
## 9      Japan  48600.22
## 10     Belgium  47858.02
## 11     Sweden  43652.09
## 12     Norway  40283.20
## 13     Portugal  32679.45
## 14     Finland  23306.79
## 15     Channel Islands  22059.72
## 16     Denmark  21291.79
## 17     Italy  21065.45
## 18     Cyprus  16829.62
## 19     Singapore  11497.98
## 20     Poland  10264.54
## 21     Austria  9883.01
## 22     Israel  9088.79
## 23     Greece  6128.80
## 24     Iceland  5438.44
## 25     Canada  4315.90
## 26     USA  4138.83
## 27     Unspecified  2952.93
## 28     Malta  2890.34
## 29 United Arab Emirates  2747.72
## 30     Lebanon  2085.13
## 31     Lithuania  1804.34
## 32 European Community  1575.65
## 33     Brazil  1225.35
## 34     RSA  1090.43
## 35     Czech Republic  1026.26
## 36     Bahrain  716.78
## 37     Saudi Arabia  181.44
```

```
# summary
# split the Excel file into 3 datasets
# cleaned nulls/duplicates/invalid values
# joined transactions+products (inner) with customers (left)
# computed Revenue, aggregated top products/countries/customers with group_by/summarise
# segmented customers with case_when
# saved everything to SQLite and queried it back to confirm.

# business insights from output
# revenue is heavily concentrated in a single market - United Kingdom, at over 82% of total sales.
# total revenue was 10,752,840, and the UK alone contributed 8,861,857.
# the business is effectively a UK native with less international outreach.
```

this signals limited geographic diversification.

the top 5 customers alone generated 1,303,492 - about 12% of total revenue

customer 18102 contributed 408,760 (nearly 4% of all revenue) individually.

loss of such customers could lead to an obvious collapse of the business

Saudi Arabia is the weakest market, and the gap suggests that it is a reachability issue and not a demand issue

this is because the major contributors are in proximity of UK and Europe